

**3** An object of mass 20 kg moves along a straight line on a smooth horizontal surface. A force  $F$  acts on the object in its direction of motion. A graph of  $F$  against time  $t$  is shown below.

If the velocity of the object at  $t = 4.0$  s is  $4.0 \text{ m s}^{-1}$ , what is its velocity at  $t = 6.0$  s?

**A**  $3.0 \text{ m s}^{-1}$

**B**  $3.3 \text{ m s}^{-1}$

**C**  $4.7 \text{ m s}^{-1}$

**D**  $5.0 \text{ m s}^{-1}$

**L2 Answer: C**

Area underneath the  $F - t$  graph gives the **change in** momentum of the object.

$$\text{Between } 4.0 \text{ s to } 6.0 \text{ s, area} = \frac{1}{2} (6 - 4) \left( \frac{2}{3} (20) \right) = 13.333 \text{ Ns}$$

Let positive direction be the direction of the object's motion.

Since  $F$  acts in the direction of motion, it acts in the positive direction, and the change in momentum is also of positive direction, i.e.  $\Delta p = +13.333 \text{ Ns}$ .

$$\text{Final momentum} - \text{Initial momentum} = +13.333$$

$$m(v_f - v_i) = +13.333$$

$$(20)(v_f - (+4.0)) = +13.333$$

$$v_f = +4.7 \text{ m s}^{-1}$$

[Turn over

|   |                                                                                                             |
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| 4 | Water is ejected at a speed of $0.5 \text{ m s}^{-1}$ onto a wall from the nozzle of a hose with a diameter |
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